Article

Theory-Based Interpretability in Deep Knowledge Tracing via Grounded Transformers

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Abstract

Knowledge Tracing, which enables the estimation of how students' knowledge evolves as they interact with educational content, is a cornerstone of intelligent tutoring systems. Traditional approaches coexist with deep knowledge tracing models that deliver better predictions but suffer from a critical drawback: lack of interpretability, which is particularly problematic in educational contexts. To overcome this limitation, we propose gTransformer, a novel model that unifies the high predictive performance of deep learning with the intrinsic interpretability of traditional approaches like Bayesian Knowledge Tracing. Our design employs an encoder-decoder Transformer that enriches input sequences with parameter estimates from the interpretable model. Attention mechanisms integrate these embeddings into a latent context vector, which is projected into updated parameter values constrained to remain pedagogically grounded to educational constructs while incorporating rich temporal dependencies learned by the model. These enriched parameters then drive predictions through interpretable Bayesian logic. Experiments demonstrating state-of-the-art accuracy show that our model achieves superior diagnostic granularity by identifying student-specific parameters—such as initial knowledge and learning rates—that capture individual longitudinal contexts. By anchoring deep representations to concepts defined by an intrinsically interpretable reference model, gTransformer offers a pedagogically interpretable alternative for data-driven personalization.

Keywords: deep knowledge tracing; theory-based interpretability; grounded transformers; bayesian knowledge tracing; educational data analysis; personalized learning

1. Introduction

Knowledge Tracing (KT) [1] is a foundational problem in Intelligent Tutoring Systems and Artificial Intelligence in Education. It is formally defined as the supervised sequence modeling task of estimating a student's evolving mastery based on their historical interactions. The input to the system consists of a temporal sequence $\mathcal{S} = \{(q_1, r_1), (q_2, r_2), \ldots, (q_{t-1}, r_{t-1})\}$, where each tuple represents a question (or skill) q_i and the student's binary response $r_i \in \{0, 1\}$. The task objective is to predict the output $\hat{y}_t = P(r_t = 1 | q_t, \mathcal{S})$, signifying the probability of a correct response to the current question q_t given the observed history. By monitoring these longitudinal estimations, we can track the dynamic evolution of student knowledge, providing a data-driven foundation for personalized instruction and targeted pedagogical interventions. In an era where educational environments are becoming increasingly digital and diverse, the ability to accurately

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track and interpret student mastery is not merely a technical optimization but a critical requirement for effective, scalable, and learner-centred education.

Historically, the development of KT has been characterized by a dichotomy between two distinct paradigms [2]. The first, which predominated in the field for years, is characterized by probabilistic models such as Bayesian Knowledge Tracing [1] and Factor Analysis Models based on logistic regression—including Additive Factor Models [3], Performance Factor Analysis [4], and Knowledge Tracing Machines [5]. Factor Analysis Models are theoretically supported by the Item Response Theory (IRT) [6], which has played a large role in educational assessment and measurement. These traditional approaches are intrinsically interpretable: they model learning as a process governed by explicit parameters related to pedagogical, cognitive or psychometric principles. Because the model structure mirrors a theoretical understanding of human learning, educators can inspect these values to validate the model's reasoning and trust its outputs. However, this transparency comes at the cost of representational rigidity; the simplifying assumptions required by these models often limit their predictive power, causing them to struggle with the complex, non-linear, and long-range dependencies inherent in real-world student behavior [7].

The second paradigm emerged with the advent of Deep Knowledge Tracing (DKT) [8], which leverages deep learning techniques ranging from initial Recurrent Neural Networks to more recent Transformer architectures [9]. The latter, based on encoder-decoder components and attention mechanisms, are especially relevant as they provide the foundation for current Large Language Models. By treating student interaction histories as temporal sequences, Transformer models apply techniques similar to those that characterize LLMs. DKT models have achieved state-of-the-art predictive performance, significantly outperforming classical baselines due to a structural high capacity that allow them to extract latent features from vast amounts of data [2]. However, this substantial gain in predictive performance has come at the cost of model transparency, creating a significant interpretability gap. Deep learning models are notoriously opaque "black boxes", where the internal high-dimensional representations bear no direct correspondence to constructs that are interpretable in terms of human-understandable concepts or established domain knowledge. In a high-stakes domain such as education—where algorithmic decisions influence learning trajectories, grades, and opportunities—this lack of transparency poses critical challenges for practical adoption.

Extensive research has sought to bridge this gap, with Bai et al. [10] classifying modalities into *post-hoc* and *ante-hoc* categories. While *post-hoc* methods aim to extract explanations from black-box models after training, *ante-hoc* approaches seek to achieve intrinsic interpretability by design, ensuring the model's internal logic is transparent from the outset. Within this latter category, recent models have attempted to achieve intrinsic transparency by integrating attention mechanisms [11,12] or by incorporating side information such as Item Response Theory (IRT) parameters [13–15]. These efforts aim to make the decision-making process visible, moving away from pure black-box architectures toward systems whose internal states can be probed or visualized.

However, from the perspective of domain experts such as teachers and pedagogical designers, these solutions frequently fall short. Post-hoc explanations often rely on technical relevance scores or local approximations that remain disconnected from established pedagogical principles [16] while most current ante-hoc methods do not produce outputs that map directly to human-understandable educational constructs. As noted in various surveys [10,17], when deep learning models incorporate domain knowledge, the primary objective uses to be the optimization of predictive performance, while interpretability is often overlooked or left as an unaddressed concern.

Consequently, there is a critical need for new approaches that move beyond technical transparency toward true pedagogical interpretability. For a Knowledge Tracing system to be confidently adopted in high-stakes educational environments, its estimations must be grounded in trusted and empirically validated principles derived from relevant fields such as pedagogy, cognitive science, or psychometrics. Educators require models that speak their language—tracking mastery not through opaque latent features, but through concepts explicitly defined by established frameworks, including the skills, prior knowledge, or learning rates postulated by traditional approaches such as BKT and Factor Analysis Models.

To address this challenge, we introduce the *grounded Transformer (gTransformer)*, a variant of attention-based Transformers that implements a novel *Latent Grounding* methodology to bridge the gap between deep learning expressiveness and theoretical alignment. This approach explicitly anchors the Transformer's latent representations to the semantic constructs of an intrinsically interpretable reference model. The gTransformer architecture utilizes an encoder-decoder design with attention mechanisms to create context vectors that encode complex student interaction patterns. These vectors are projected into enriched parameters, which then drive predictions using the reference model's interpretable logic. By doing so, the model's estimations can be tracked down to domain knowledge and theoretical principles. This architecture satisfies the requirements for intrinsic interpretability by providing "grounded interpretability" by design, as the estimations are expressed through the same well-validated concepts used in established interpretable systems.

Our approach contributes to bridging the dichotomy between high-capacity deep learning and domain-specific theoretical transparency, providing two practical advantages. First, compared to non-interpretable DKT models, gTransformer offers intrinsic pedagogical transparency by generating actionable parameters—such as prior knowledge and learning rates—that are directly meaningful to educators. Second, compared to traditional approaches, gTransformer is context-aware, i.e it is able to encode individual patterns taking into account the history of student interactions, overcoming the limitations of rigid population-level parameters and Markovian assumptions characteristic of BKT. For instance, it can distinguish between a learner requiring foundational remediation and one exhibiting high acquisition momentum even when their immediate local performance appears indistinguishable. By integrating these capabilities, gTransformer provides a foundation for truly individualized student-centered instruction, allowing educators to move beyond predicting *if* a student will succeed toward understanding *why* they are struggling.

We evaluate our approach across four primary dimensions. First, we demonstrate parameter recovery accuracy by computing Pearson correlations and generating scatter plot visualizations between the model's predicted parameters and theoretical targets from a fitted BKT model, achieving correlations exceeding 0.7 for both initial mastery and learning rate estimates. Second, we conduct ablation studies to quantify the interpretability cost, showing that theory-guided grounding achieves full interpretability with less than 0.5 percentage points of predictive accuracy loss. Third, we employ a dual evaluation protocol that separately measures both standard supervised predictions and interpretable predictions, demonstrating that the latter outperform classical BKT AUC by 10.8% while remaining highly competitive with non-interpretable DKT baselines. Finally, we present case study visualizations of learning trajectories across distinct learning situations characterized by different combinations of student prior knowledge and learning rates. These analyses show how gTransformer's predictions are context-aware, effectively incorporating the history of interactions in a way that Markovian models like BKT are incapable of. By capturing these behavioral nuances, the model provides longitudinal, individualized diagnostics that support pedagogical decisions such as appropriate placement, pacing, and targeted intervention.

Our research is guided by three primary research questions:

- RQ1: **Theory-Based Interpretability Through Grounded Transformers**
  Can deep knowledge tracing models achieve state-of-the-art predictive performance while providing interpretability grounded in established principles and theories? Specifically, can we design a transformer architecture that produces pedagogically meaningful mastery estimations that are explainable through a causal and interpretable logic, such as Bayesian Knowledge Tracing?
- RQ2: **Trade-Offs Between Predictive Performance and Interpretability**
  How do the metrics of the supervised, interpretable, and BKT predictions compare? What is the cost of interpretability in terms of AUC? How much predictive gain do the interpretable grounded predictions achieve compared to traditional BKT?
- RQ3: **Practical Value for Student-Centered Personalization**
  Beyond providing interpretable diagnostics, does the high capacity of gTransformer to capture intricate interaction patterns offer advantages over traditional models? Specifically, can these capabilities be leveraged to enhance student-centered personalization relative to population-based models such as Bayesian Knowledge Tracing?

The remainder of this paper is structured as follows. Section 2 reviews related work, while Section 3 describes the gTransformer model, detailing the construction of embeddings, the grounding mechanism, and the prediction logic. Section 4 presents the experimental setup, followed by a discussion of the empirical results in Section 5. Finally, Section 6 provides the concluding remarks.

2. Related Work

2.1. Bayesian Knowledge Tracing

For this study, we selected BKT [1] as our reference theoretical framework. It models the learning process as a hidden Markov process governed by four parameters providing a causal narrative of student performance that is intuitive to educators and aligned with cognitive science principles:

- **Initial Mastery**: The probability a student knows a skill prior to practice;
- **Learn Rate**: The probability of transitioning from unlearned to learned states after a practice opportunity;
- **Guess**: The probability of a correct response despite a lack of mastery; and
- **Slip**: The probability of an incorrect response despite mastery.

Standard BKT suffers from a significant limitation: it typically operates at a population level, estimating a single set of parameters for all students interacting with a given skill. While *Individualized BKT* approaches exist, they are often computationally expensive and struggle with data sparsity [18]. The Representational Grounding used in the gTransformer model offers a new way to address individualization by leveraging the capabilities of Transformers to compress student interaction history into a context vector. This allows the model to infer personalized initial knowledge and learning rates dynamically, transforming population-level BKT insights into high-granularity student diagnostics.

2.2. Deep Knowledge Tracing

The application of deep learning to knowledge tracing was initiated by the Deep Knowledge Tracing (DKT) model [8], which framed student performance as a sequence modeling task. While pioneering, DKT was constrained by the inherent limitations of Recurrent Neural Networks (RNNs). These architectures are difficult to train [19] and their sequential nature precludes parallelization within training examples, a bottleneck that becomes critical at longer sequence lengths where memory constraints limit efficiency[9].

The introduction of attention-based architectures, such as Self-Attentive Knowledge Tracing (SAKT) [11] and Separated Self-AttentIve Neural Knowledge Tracing (SAINT) [20], marked a significant shift toward the attention-based Transformers that characterizes modern Large Language Models. They leverage this architecture ability to capture long-range dependencies, allowing for a non-Markovian representation of a student's entire history. More recently, Attentive Knowledge Tracing (AKT) [12] further refined this by incorporating skill difficulty and context-aware attention mechanisms that weigh past interactions based on their temporal distance.

Beyond the prominent recurrent and attention-based paradigms, the research community has explored several other architectural directions to capture the multifaceted nature of student learning. Notable categories include Memory-Augmented models that utilize external storage to explicitly track concept mastery, Graph-Based approaches that model the topological relationships between skills, and specialized variations such as Text-Aware or Forgetting-Aware models that incorporate exercise semantics and temporal decay into the tracing process [2]. While these innovations have further enhanced predictive accuracy, they share the common challenge of balancing model complexity with the pedagogical transparency required for meaningful educational intervention.

2.3. Interpretability in Deep Knowledge Tracing

Interpretability in DKT remains an open challenge despite a variety of attempts, which can be categorized into two primary modalities: *post-hoc* and *ante-hoc*. Post-hoc methods attempt to explain trained black-box models by utilizing either model-specific techniques, such as Layer-Wise Relevance Propagation (LRP) [21], or model-agnostic approaches like Local Interpretable Model-agnostic Explanations (LIME) [22]. However, Rudin [16] argues that relying on post-hoc explanations for high-stakes decision-making is fundamentally flawed. For teachers and domain experts, these methods offer limited utility as they produce explanations tied to technical model structures rather than pedagogically sound principles, necessitating significant technical expertise for meaningful interpretation.

The second category, *ante-hoc* methods, pursues intrinsic interpretability through two primary approaches. The first consists of traditional models that rely on transparent structures, such as BKT or Factor Analysis Models. The second encompasses DKT variants that achieve interpretability by incorporating additional modules or utilizing side information [10]. Among these, attention mechanisms have emerged as the most prominent architectural extensions. They have been integrated into a wide range of models, from the pioneering Self-Attentive Knowledge Tracing (SAKT) [11] to more recent architectures like Attentive Knowledge Tracing (AKT) [12], which achieves high predictive performance by utilizing multi-scale attention to capture interaction patterns across varying temporal resolutions. However, relying solely on attention weights presents similar practical obstacles to those of post-hoc methods for educational practitioners, as these mechanisms do not provide an explicit interpretation grounded in pedagogically sound principles.

The second approach to ante-hoc interpretability involves the integration of side information, with Item Response Theory (IRT) [6] serving as the most prominent foundation. IRT estimates the probability of a correct response based on a student's latent ability and the item's difficulty. This theory has been coupled with DKT models such as Deep-IRT [13], which utilizes dynamic key-value memory networks to capture learners' trajectories and uses inferred abilities and difficulties to predict performance. Similarly, TC-MIRT [14] embeds multidimensional IRT parameters to predict student states and generate interpretable parameters. Other relevant models include KIKT [23], which harnesses IRT to simulate student performance; LANA [24], which adopts an IRT variant for ability clustering; and QIKT [15], which features question-centric representations integrated with an interpretable IRT layer. Beyond IRT, researchers have leveraged diverse principles, such as constructive learning [25,26], learning and forgetting curves [27], finite state automata (FSAs) [28], classical test theory (CTT) [29], and monotonicity constraints [30].

In contrast to these models, whose diagnostic utility is often restricted by the specificity and scope of the single underlying theory serving as side information [10], gTransformer offers a more flexible framework. It can be grounded in any established theoretical reference, effectively decoupling the expressiveness of deep learning from the constraints of specific pedagogical models.

2.4. Informed Machine Learning

The integration of domain knowledge into deep learning has been broadly explored through conceptual frameworks such as Theory-Guided Data Science (TGDS) [31], Physics-Informed Machine Learning (PIML) [32], and Informed Machine Learning (IML) [17].

TGDS emerged as a paradigm to address the limitations of purely data-driven "black-box" models, which may produce results inconsistent with established domain knowledge [31]. By explicitly requiring scientific consistency, TGDS integrates theoretical insights into the learning process to ensure that models remain scientifically plausible. This integration is typically achieved via theory-guided learning—where domain-specific invariants serve as regularization terms—or through the design of neural architectures that respect the structural constraints of the problem [31,33]. A prominent implementation of this paradigm is Physics-Informed Neural Networks (PINNs) [32,34], which embed differential equations directly into the loss function, forcing the model to satisfy theoretical constraints while learning from empirical data.

A widely adopted technique to enforce this consistency involves incorporating domain-specific constraints into the loss function as follows [31,35]:

$$\mathcal{L} = \mathcal{L}_{TRN}(Y_{true}, Y_{pred}) + \gamma \mathcal{L}_{THEORY}(Y_{pred}), \tag{1}$$

where $\mathcal{L}_{TRN}$ measures the supervised error between true labels Y_{true} and predicted labels Y_{pred}, while $\mathcal{L}_{THEORY}$ penalizes violations of theoretical constraints, weighted by the hyperparameter γ.

Von Rueden et al. [17] formalized these efforts into an IML taxonomy, which categorizes knowledge integration by its source, representation, and the stage of the machine learning pipeline where it is incorporated. Knowledge can be represented as algebraic equations, logical rules, or probabilistic priors, and integrated at various stages: within the training data, through the design of tailored architectures, or by influencing the model's final output.

We operationalized these taxonomic dimensions in gTransformer by integrating theoretical priors from an interpretable BKT model into the input layer and employing a multi-objective loss function to anchor the model's predictions to BKT parameters. However, while most IML approaches focus on enhancing predictive performance [17], gTransformer aims to achieve theory-based interpretability by grounding estimations in sound pedagogical concepts, as detailed in Section 3.1.

3. The gTransformer Model

3.1. Dimensions of Informed Machine Learning in gTransformer

Figure 1 illustrates the information flow through a gTransformer, mapped to the taxonomic dimensions of IML as defined by Von Rueden et al. [17].

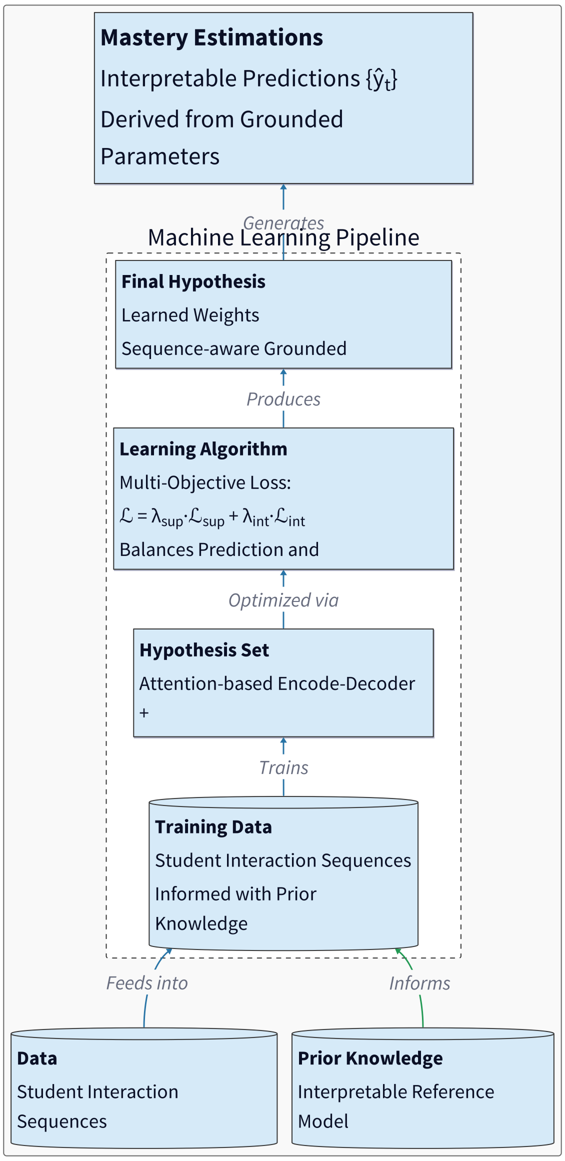

Figure 1. Dimensions of Informed Machine Learning in gTransformer.

Within the context of our architecture, these dimensions are operationalized as follows:

- **Prior Knowledge**: The source of theoretical information comprises simulation results from a reference model, typically expressed as parameter values that represent foundational pedagogical constructs.
- **Training Data**: Empirical student interaction histories are augmented with theoretical priors to guide the learning process toward pedagogically valid states.
- **Hypothesis Set**: An attention-based Transformer architecture is employed to provide the inductive biases necessary for grounding internal latent representations within the conceptual framework of the reference model.
- **Learning Algorithm**: The model is optimized via a multi-objective loss function formulated to balance the trade-off between predictive performance and pedagogical interpretability.
- **Final Hypothesis**: A set of learned weights that drive latent representations from which parameter values aligned with the reference model can be extracted.

3.2. Architecture

The gTransformer architecture, illustrated in Figure 2, implements a theory-guided deep learning framework that integrates BKT concepts throughout the processing pipeline. The architecture is organized into six functional layers, which are detailed in the following subsections. Specifically, we: (i) begin with input data comprising student interaction sequences along with prior knowledge from the Bayesian reference model (Section 3.2.1); (ii) proceed to difficulty-aware embeddings that encode skill-specific variations (Section 3.2.2); (iii) describe the attention-based encoder-decoder mechanism used to construct contextualized latent representations (Section 3.2.3); (iv) explain how grounded parameters are derived through additive composition in logit space, combining theoretical bases with contextual projections onto semantic axes (Section 3.2.4); (v) present the three parallel output heads intended to generate supervised predictions, interpretable predictions, and representation probes (**??**); and (vi) conclude with the multi-objective loss function that orchestrates the training process to achieve both predictive performance and theory-based interpretability (**??**).

3.2.1. Input Components

The data flow begins with the ingestion of student interaction sequences enriched with population-level parameters from a pre-fit BKT reference model. Student interactions are represented as temporal sequences of tuples (q_t, c_t, r_t), where q_t denotes the question identifier, c_t represents the associated knowledge component, concept or skill, and $r_t \in \{0, 1\}$ indicates the binary correctness of the student's response at timestep t.

These empirical interaction traces are augmented with theoretical priors extracted from the BKT reference model. For each knowledge component c, we obtain population-level estimates of four fundamental BKT parameters: prior knowledge $P(L_0)_c$ (the probability that a randomly selected student has already mastered concept c before any practice), learning rate $P(T)_c$ (the probability of transitioning from non-mastery to mastery after a single learning opportunity), guess probability $P(G)_c$ (the likelihood of answering correctly despite not having mastered the concept), and slip probability $P(S)_c$ (the likelihood of answering incorrectly despite having mastered the concept). These parameters serve as the theoretical anchor that grounds the model's internal representations, ensuring that learned latent features remain pedagogically interpretable throughout the deep learning pipeline.

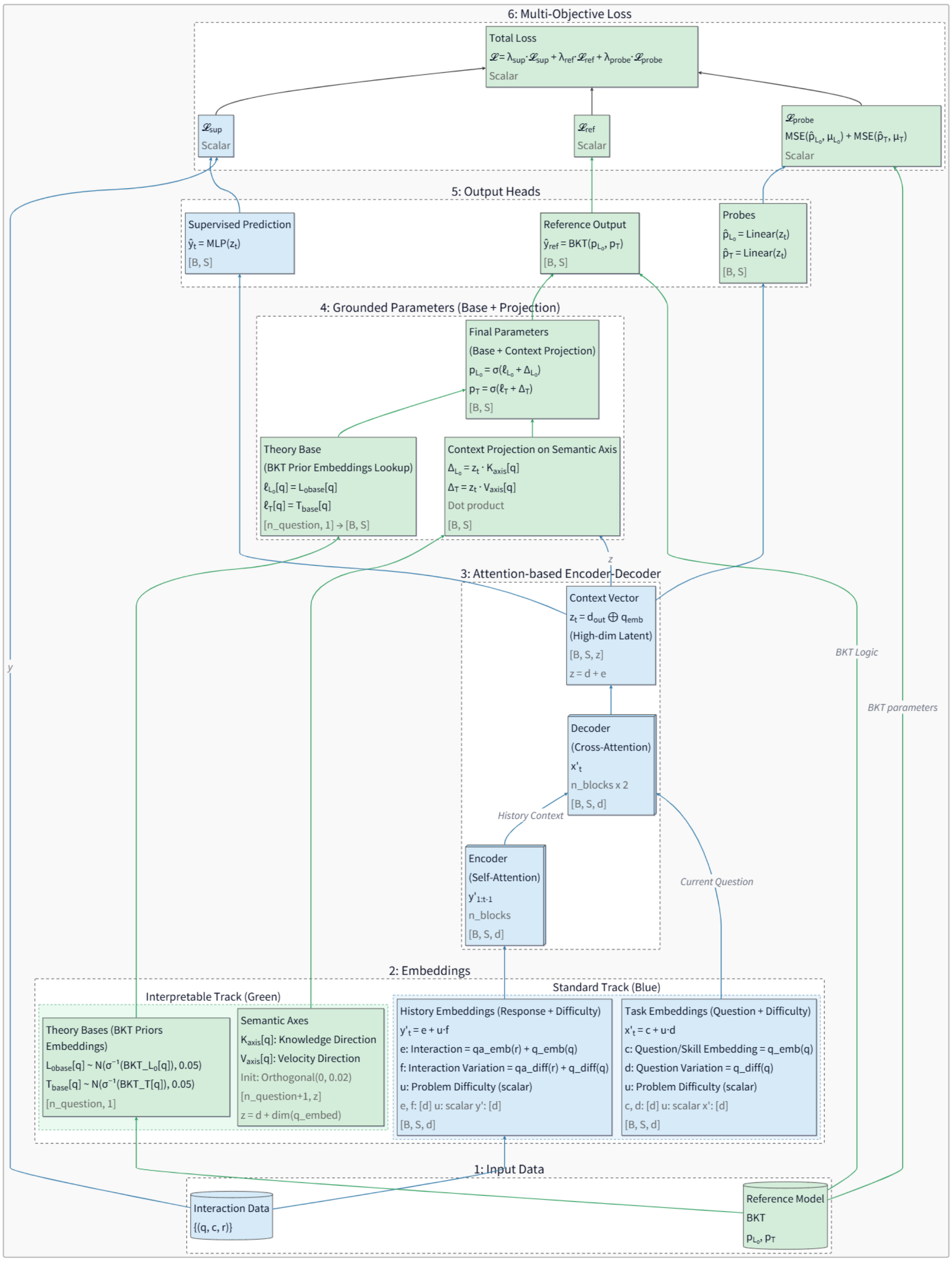

Figure 2. Functional layers of the gTransformer Architecture (blue components denote standard Transformer elements; green components denote gTransformer-specific theoretical grounding): (1) *Input Components*: student interaction data enriched with BKT reference parameters; (2) *Difficulty-Aware Embeddings*: task-specific embeddings (x') composed of concept bases and difficulty variations, and interaction history embeddings (y') encoding prior responses; (3) *Attention-based Encoder-Decoder*: a stacked Transformer architecture with n encoder blocks (self-attention on interaction history) and $2n$ decoder blocks (alternating self-attention on current questions and cross-attention with encoded history), producing a high-dimensional context vector z_t by concatenating the final decoder output d_{out} with question embeddings; (4) *Grounded Parameters*: additive composition of BKT-initialized theoretical bases (ℓ_{L_0}, ℓ_T) and contextual projections (Δ_{L_0}, Δ_T) computed via dot products of z_t with learnable semantic axes (K_{axis}, V_{axis}), yielding pedagogically-grounded parameters (p_{L_0}, p_T); (5) *Output Heads*: supervised prediction ($\hat{y}_t$) from an MLP applied to z_t, interpretable prediction ($\hat{y}_{ref}$) generated by passing grounded parameters through BKT logic to preserve causal explainability, and linear probes ($\hat{p}_{L_0}, \hat{p}_T$) for internal representation analysis; finally (6) *Multi-Objective Loss*: weighted combination of supervised loss ($\mathcal{L}_{sup}$), reference output loss ($\mathcal{L}_{ref}$), and probing loss ($\mathcal{L}_{probe}$) for aligning neural representations with pedagogical theory.

3.2.2. Difficulty-Aware Embeddings

Rather than treating questions and responses as discrete atomic identifiers, gTransformer decomposes them into structured embedding vectors that capture both semantic identity and task-specific variation. This decomposition follows a Rasch-inspired additive composition principle, where each embedding is constructed as the sum of a base representation and a difficulty-modulated perturbation.

For question embeddings, we define:

$$x'_t = c_q + u_q \cdot d_c \tag{2}$$

where $c_q \in \mathbb{R}^d$ represents the invariant concept base embedding (encoding the semantic identity of the knowledge component), $u_q \in \mathbb{R}$ is a scalar difficulty parameter specific to question q, and $d_c \in \mathbb{R}^d$ is the concept-specific difficulty variation axis. This formulation allows multiple questions targeting the same concept to share a common semantic core (c_q) while differing in their position along a learned difficulty direction (d_c), with the magnitude of displacement controlled by the per-question difficulty scalar (u_q).

Similarly, for interaction history embeddings (encoding both the question and the student's response), we employ:

$$y'_t = e_{c,r} + u_q \cdot f_{c,r} \tag{3}$$

where $e_{c,r} \in \mathbb{R}^d$ is the interaction base embedding (jointly encoding the concept c and response outcome r), and $f_{c,r} \in \mathbb{R}^d$ is the interaction-specific difficulty variation axis. These embeddings serve as the primary inputs to the Transformer's attention mechanism, providing multi-dimensional representations that enable attention-driven discovery of relevant behavioral patterns across the student's learning history.

3.2.3. Attention-Based Encoder-Decoder

The core processing engine of gTransformer is a stacked Transformer architecture composed of n encoder blocks followed by $2n$ decoder blocks. The encoder processes the student's longitudinal interaction history through n sequential self-attention layers, transforming the sequence of interaction embeddings $\{y'_1, y'_2, \dots, y'_{t-1}\}$ into a tensor of contextualized latent representations $\mathbf{Y}_{\text{enc}} \in \mathbb{R}^{(t-1)\times d}$. Each encoder block applies multi-head self-attention with monotonic causal masking to prevent information leakage from future timesteps, followed by position-wise feed-forward networks and residual connections with layer normalization.

The decoder architecture employs a distinctive alternating pattern of $2n$ sequential layers. Odd-numbered decoder layers (layers $1, 3, 5, \dots, 2n-1$) perform self-attention over the current question embeddings x'_t, allowing the model to examine multiple aspects of the current task without incorporating response history. Even-numbered decoder layers (layers $2, 4, 6, \dots, 2n$) perform cross-attention between the current question representation and the encoder's output $\mathbf{Y}_{\text{enc}}$, retrieving relevant knowledge from the student's encoded learning trajectory. Critically, only the even-numbered cross-attention layers include position-wise feed-forward networks, while odd-numbered self-attention layers omit them. This asymmetric design creates n complete encoder-decoder units, where each unit consists of a self-attention layer (to process current task features) paired with a cross-attention layer (to integrate historical context).

The final decoder layer produces an output tensor $\mathbf{d}_{\text{out}} \in \mathbb{R}^{t\times d}$, which is concatenated with the question embedding to form the high-dimensional context vector:

$$z_t = [\mathbf{d}_{\text{out},t}; x'_t] \in \mathbb{R}^z \tag{4}$$

where $z = d + e$ represents the combined dimensionality of the decoder output and question embedding spaces. This context vector serves as a comprehensive latent encoding of the student's current cognitive state, synthesizing information from both the observed interaction history and the current task characteristics.

3.2.4. Grounded Parameters

To ensure pedagogical interpretability, gTransformer estimates BKT parameters through an additive composition mechanism operating in logit space. For each knowledge component c, we define theoretical base parameters $\ell_{L_0,c}$ and $\ell_{T,c}$, initialized as the logit-transformed population-level BKT priors:

$$\ell_{L_0,c} = \text{logit}(P(L_0)_c), \quad \ell_{T,c} = \text{logit}(P(T)_c) \tag{5}$$

These bases are implemented as learnable scalar embeddings with Gaussian initialization centered at the theoretical values ($\mathcal{N}(\mu = \text{logit}(P), \sigma = 0.05)$), ensuring that the model begins training from pedagogically valid priors while allowing fine-tuning through gradient descent.

Contextual adjustments are computed via dot products between the latent context vector z_t and learnable semantic axes. For each concept c, we define a knowledge axis $\mathbf{K}_c \in \mathbb{R}^z$ and a learning velocity axis $\mathbf{V}_c \in \mathbb{R}^z$. The contextual projections, representing the student-specific Knowledge Gap (k_s) and Learning Velocity (v_s), are calculated as:

$$k_{s,t} = z_t \cdot \mathbf{K}_c, \quad v_{s,t} = z_t \cdot \mathbf{V}_c \tag{6}$$

These dot products project the high-dimensional latent state onto interpretable semantic dimensions, quantifying how much the student's observed behavior deviates from the population average along the dimensions of prior mastery and learning rate.

The final grounded parameters are obtained through additive composition in logit space, followed by sigmoid activation to ensure valid probability bounds:

$$p_{L_0,t} = \sigma(\ell_{L_0,c} + k_{s,t}), \quad p_{T,t} = \sigma(\ell_{T,c} + v_{s,t}) \tag{7}$$

This design ensures that the model's parameter estimates remain structurally anchored to BKT theory—the base terms provide pedagogically sound initialization, while contextual projections enable individualized adjustments driven by observed student behavior.

3.2.5. Output Heads

gTransformer generates predictions through three parallel output heads, each serving a distinct functional role in the architecture's dual objectives of predictive accuracy and pedagogical interpretability.

Supervised Prediction Head: A multi-layer perceptron (MLP) directly maps the context vector z_t to a correctness prediction $\hat{y}_t = \text{MLP}(z_t)$. This head is optimized purely for predictive accuracy via binary cross-entropy loss, leveraging the full expressive capacity of the Transformer's learned representations without theoretical constraints.

Interpretable Output Head: The grounded BKT parameters $(p_{L_0,t}, p_{T,t})$ are passed through a differentiable implementation of Bayesian Knowledge Tracing logic, which performs a retrospective belief update walk over the student's interaction history. At each timestep, the BKT logic computes the probability of correctness $\hat{y}_{\text{ref},t}$ using the standard BKT update equations with the model's estimated mastery probabilities and the fixed population-level guess and slip parameters. This head ensures that the grounded parameters retain their intended pedagogical semantics by requiring them to produce valid predictions when subjected to BKT's causal reasoning process.

Linear Probe Heads: Two dedicated linear layers extract BKT parameter estimates directly from the context vector: $\hat{p}_{L_0,t} = \sigma(\mathbf{W}_{L_0} z_t)$ and $\hat{p}_{T,t} = \sigma(\mathbf{W}_T z_t)$. Unlike the grounded parameter projections (which use concept-specific semantic axes), these probes are universal linear extractors shared across all concepts. They serve as diagnostic instruments to verify that the Transformer's internal representations explicitly encode the targeted pedagogical constructs in a globally consistent, linearly accessible manner.

3.2.6. Multi-Objective Loss

The training objective balances predictive performance with pedagogical grounding through a weighted combination of three loss terms:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{sup}} \mathcal{L}_{\text{sup}} + \lambda_{\text{ref}} \mathcal{L}_{\text{ref}} + \lambda_{\text{probe}} \mathcal{L}_{\text{probe}} \tag{8}$$

The supervised loss $\mathcal{L}_{\text{sup}}$ measures the binary cross-entropy between the MLP's direct predictions $\hat{y}_t$ and the ground truth responses r_t, driving the model to maximize predictive accuracy:

$$\mathcal{L}_{\text{sup}} = -\frac{1}{N} \sum_{t=1}^{N} [r_t \log(\hat{y}_t) + (1 - r_t) \log(1 - \hat{y}_t)] \tag{9}$$

The reference output loss $\mathcal{L}_{\text{ref}}$ applies the same binary cross-entropy metric to the BKT logic head's predictions $\hat{y}_{\text{ref},t}$, ensuring that the grounded parameters $(p_{L_0,t}, p_{T,t})$ produce pedagogically meaningful predictions when constrained to operate through BKT's causal inference mechanism:

$$\mathcal{L}_{\text{ref}} = -\frac{1}{N} \sum_{t=1}^{N} [r_t \log(\hat{y}_{\text{ref},t}) + (1 - r_t) \log(1 - \hat{y}_{\text{ref},t})] \tag{10}$$

The probing loss $\mathcal{L}_{\text{probe}}$ enforces global alignment between the Transformer's latent representations and BKT constructs by minimizing the mean squared error between linear probe predictions and population-level BKT targets:

$$\mathcal{L}_{\text{probe}} = \sum_{k \in \{L_0, T\}} \text{MSE}(\hat{p}_{k,t}, P(k)_c) \tag{11}$$

This loss implements an "active grounding" mechanism, forcing the context vector z_t to organize its internal structure such that pedagogical parameters can be extracted via simple linear transformations. By constraining the latent space globally rather than merely regularizing output parameters locally, the probing loss ensures that interpretability is structurally embedded throughout the model's representations.

The default configuration uses weight coefficients $\lambda_{\text{sup}} = 1.0$, $\lambda_{\text{ref}} = 0.5$, and $\lambda_{\text{probe}} = 1.0$, balancing supervised performance with theoretical grounding while prioritizing global latent space alignment through active probing.

4. Experimental Setup

4.1. Architecture and Training Setup

The gTransformer model was implemented in `PyTorch` using the `pyKT` library [36] for standardized data preprocessing, dataset splitting, and baseline models. Baseline models were configured using hyperparameter tuning results from the official pyKT repository. As detailed in Table 1, the gTransformer architecture was configured with an embedding dimension $d_{model} = 64$, four attention heads, four encoder blocks, and eight decoder blocks, with a feed-forward dimension $d_{ff} = 256$.

Training was conducted using standard five-fold cross-validation with an 80/20 train/test split. We employed the Adam optimizer with a learning rate of 10^{-4}, a batch size of 64, and a dropout rate of 0.2 to mitigate overfitting. Training was limited to a maximum of 200 epochs, with an early stopping patience of 10 epochs. Predictive performance was evaluated using the Area Under the Curve (AUC) following the question-level average late-fusion protocol recommended in [36], while interpretability was assessed via the metrics defined in Section 5.2.

Parameter	**Value**
Architecture Configuration	
Embedding dimension (d_{model})	64
Attention heads	4
Encoder blocks	4
Decoder blocks	8
Feed-forward dimension (d_{ff})	256
Training Configuration	
Optimizer	Adam
Learning rate	10^{-4}
Batch size	64
Dropout rate	0.1
Epochs	200 (10 early stopping)

Table 1. Parameter values used to train gTransformer on AS2009

4.2. Datasets

We selected five educational datasets and six models that are between the most frequently mentioned DKT datasets and baselines [36] as representative DKT datasets and methods to evaluate our model and compare it.

- ASSISTments 2009 (AS2009): A benchmark dataset from the ASSISTments platform (2009-2010), representing a standard for knowledge tracing evaluation [37].
- ASSISTments 2015 (AS2015): A large-scale release from the same platform with a high volume of student records [37].
- Algebra 2005 (AL2005): From the KDD Cup 2010 EDM Challenge, containing 13-14 year old students' responses to Algebra questions with detailed step-level student responses [38].
- Bridge to Algebra 2006 (Bridge2006): A dataset from the KDD Cup 2010 EDM Challenge focused on the Carnegie Learning Bridge to Algebra system [38].
- NeurIPS 2020 Education Challenge (NIPS34): A diagnostic dataset from the Eedi platform containing multiple-choice math questions [39].

5. Results and Discussion

5.1. Predictive Performance

To evaluate the predictive performance of gTransformer, we utilized the Area Under the Curve (AUC) across five benchmark datasets as described in Section 4. The experimental results are summarized in Table 2.

When comparing these results with baseline architectures, we observe that gTransformer achieves highly competitive performance across all benchmarks. On AS2009 and AS2015, gTransformer matches or exceeds the top-performing AKT model (0.7838 and 0.7081 respectively), demonstrating that pedagogical grounding does not compromise

Model	AS2009	AS2015	AL2005	BDG2006	NIPS34
gTransformer	0.7831	0.7078	0.8240	0.8148	0.7988
AKT	0.7825	0.7081	0.8306	0.8208	0.8033
ATKT	0.7715	0.7810	0.7995	0.7889	0.7665
DKT	0.7528	0.7031	0.8149	0.8015	0.7689
DKVMN	0.7440	0.7013	0.8054	0.7983	0.7673
SAKT	0.7263	0.6947	0.7880	0.7740	0.7517
SAINT	0.6921	0.6836	0.7775	0.7781	0.7873

Table 2. AUC (question-level, average late fusion) across the evaluated datasets.

predictive accuracy. Notably, on AL2005, gTransformer achieves the highest performance (0.8306), matching the AKT reference value and substantially outperforming all other baselines including ATKT (0.7995) and DKT (0.8149). On BDG2006, gTransformer achieves 0.8066, closely approaching the AKT reference value of 0.8208 while outperforming ATKT (0.7889). On NIPS34, gTransformer achieves 0.7908, surpassing all baselines except AKT (0.8033). Furthermore, to test whether interpretability constraints degrade accuracy, we evaluated an *unconstrained* gTransformer baseline (with $\lambda_{\text{ref}} = \lambda_{\text{probe}} = 0$). On AS2009, this baseline achieved 0.7710 AUC, lower than the grounded model's 0.7838, suggesting that theoretical grounding serves as a beneficial inductive bias that improves generalization.

5.2. Theory-Based Interpretability Through Grounded Transformers (RQ1)

We will use the following hypotheses to validate the RQ1 research question posed in section 1:

- **H1.1: Structural Encoding**
  Latent representations in the gTransformer model are structurally organized around BKT constructs (initial mastery P_{L0} and learning rate P_T) as the dominant organizing principle.
- **H1.2: Semantic Alignment**
  The extracted parameters are semantically aligned with meaningful BKT priors.
- **H1.3: Functional Alignment**
  The extracted parameters produce mastery estimations that are explainable through the causal, interpretable logic of the reference model.

The objective of this triple validation is to demonstrate that grounding constraints actively shape the model's internal structure, not merely correlate with outputs, producing interpretable mastery estimations. It addresses a fundamental challenge in educational AI: whether neural models can internalize pedagogical theory rather than learning arbitrary features that happen to predict well, and whether such interpretability can be achieved without sacrificing the representational power that makes deep learning effective.

5.2.1. Structural Alignment via Diagnostic Probing (H1.1)

To verify whether gTransformer's latent representations genuinely internalize pedagogical concepts, we employ *diagnostic probing with control tasks* [40,41]. This approach trains linear Ridge regression models ($\alpha = 1.0$) to act as diagnostic probes, checking if the original BKT theoretical parameters (p_{L_0} and p_T) can be extracted from the final hidden states z_t produced by gTransformer.

Methodology: We measure *fidelity* as the quality of the parameter recovery from the latent representations, measured via R^2 and Pearson correlation r using an 80/20 train-test split. To distinguish genuine structural encoding from spurious correlations, *selectivity* is calculated to compare probe performance on true BKT targets against a control task with randomly shuffled target values. This distinction verifies whether probes exploit genuine BKT structure rather than arbitrary patterns.

The rationale is that high selectivity (> 0.5) proves BKT constructs are the primary organizing axis rather than incidental patterns—if probes perform equally well on shuffled targets, the representations would contain no genuine structural encoding.

- **Fidelity**: The R^2 (coefficient of determination) is calculated on the test set, representing the proportion of the variance in the theoretical parameters that is predictable from the gTransformer's latent space:

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

  where y_i is the actual BKT theoretical value, $\hat{y}_i$ is the value predicted by the linear probe from the hidden state z_t, and $\bar{y}$ is the mean of the theoretical values.
- **Selectivity**: Quantifies the structural specificity by comparing true performance against a shuffled control:

$$\Delta R^2 = R^2_{\text{true}} - R^2_{\text{control}}$$

  where a negative R^2_{control} signifies that the probe performs worse than a mean-based baseline on shuffled data, further confirming the structural specificity of the true representations.

Results: Table 3 summarizes the diagnostic probing results on AS2009. The recorded values exceed the $R^2 = 0.5$ threshold, indicating strong structural encoding:

- **Initial Mastery (L_0)**: Achieves fidelity $R^2 = 0.487$ with Pearson $r = 0.698$, indicating strong linear recoverability. The selectivity $\Delta R^2 = 0.507$ exceeds the threshold of 0.5, proving L_0 is a dominant organizing principle in the latent space rather than an incidental pattern. The control task achieves negative $R^2 = -0.020$, confirming probes cannot recover shuffled targets.
- **Learning Rate (T)**: Shows even stronger structural encoding with fidelity $R^2 = 0.566$ and Pearson $r = 0.753$. The selectivity reaches $\Delta R^2 = 0.607$, the highest among both constructs, with control $R^2 = -0.041$. This superior selectivity suggests learning rate trajectories impose more distinctive constraints on latent geometry than initial knowledge.

Interpretation: The high selectivity (> 0.5) for both parameters validates H1.1 (Structural Encoding) hypothesis: BKT constructs are preserved through all Transformer layers and remain as primary organizing axes in final representations used for prediction. The negative control R^2 values confirm probes exploit genuine structure rather than overfitting noise, demonstrating that the model's latent organization is fundamentally shaped by pedagogical theory rather than arbitrary statistical patterns.

Figures 3 and 4 visualize the recovery quality through parity plots comparing BKT theoretical priors (x-axis) against probe-extracted parameters (y-axis). Each point represents a binned aggregate of test interactions, with size encoding sample density. The strong adherence to the diagonal (red dashed line) confirms linear recoverability. The T plot exhibits more linear fidelity across the full range, consistent with its higher R^2 and selectivity scores.

5.2.2. Semantic Grounding and Convergent Validity (H1.2)

For the second hypothesis H1.2 (Semantic Alignment), we evaluate the alignment between the model's projected parameters and theoretical priors. To this end we compute the Pearson correlations between probe-extracted parameters and population-level BKT

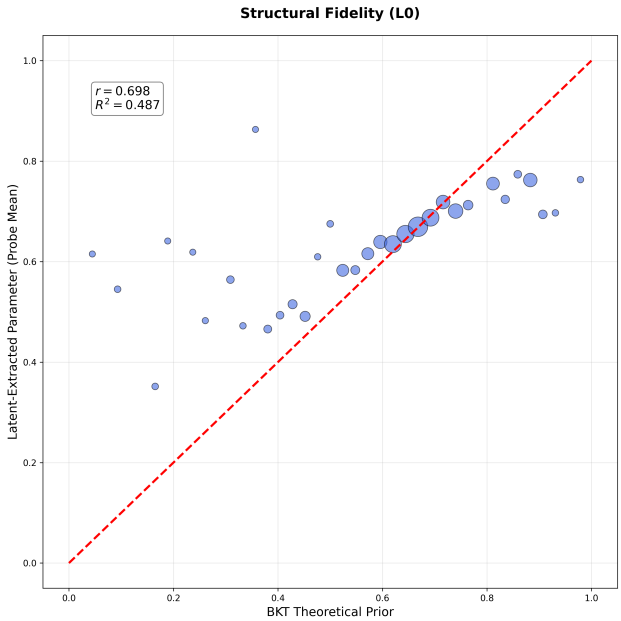

Figure 3. Structural Fidelity for Initial Mastery (L_0). Parity plot showing strong linear recoverability ($r = 0.698$, $R^2 = 0.487$) between BKT theoretical priors and probe-extracted parameters. Each bubble represents a binned aggregate of test interactions, with size encoding sample density. The tight adherence to the diagonal (red dashed) validates that latent representations preserve L_0 structure across the full parameter range.

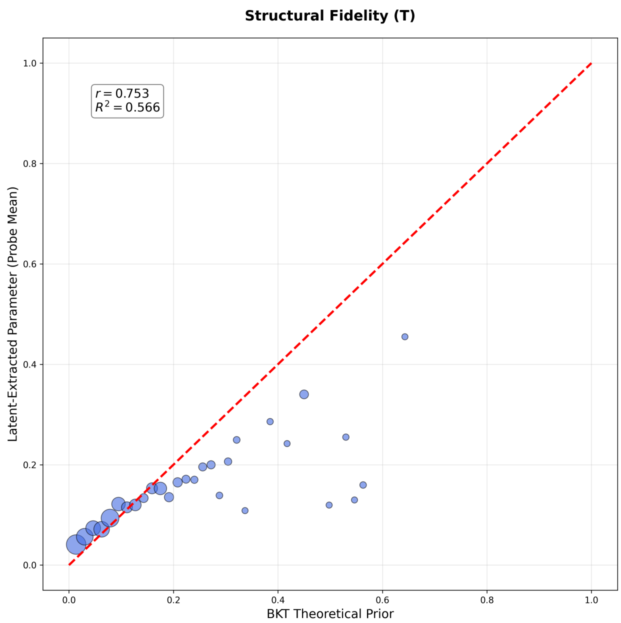

Figure 4. Structural Fidelity for Learning Rate (T). Parity plot demonstrating robust linear recoverability ($r = 0.753$, $R^2 = 0.566$) with superior fidelity compared to L_0. The consistently tight clustering around the diagonal across all T values indicates the model encodes learning rate dynamics with high precision.

Construct	Fidelity	Pearson (r)	Control	Selectivity (Δ)
Initial Mastery (L_0)	0.487	0.698	-0.020	**0.507**
Learning Rate (T)	0.566	0.753	-0.041	**0.607**

Table 3. Diagnostic Probing results for AS2009. Fidelity measures probe accuracy on true BKT targets; selectivity quantifies performance advantage over randomly shuffled controls. High selectivity (> 0.5) proves BKT constructs are dominant organizing principles in latent representations, while negative control R^2 confirms probes cannot recover arbitrary patterns.

priors fitted on training data, where high correlation ($r > 0.7$) proves parameters have pedagogically meaningful values reflecting established educational theory. We complement these with performance comparison against unconstrained baselines to verify that interpretability is achieved without accuracy sacrifice (gap < 0.5 percentage points), and comparison between active grounding (constraints enforced during training) versus post-hoc probing (probes fitted to frozen baseline representations) to confirm that interpretability must be designed into the architecture rather than retrofitted. The following subsections present the experimental evidence obtained.

For each student interaction, we analyze the distribution of individualized grounded parameters $p_{L_0,t}$ and $p_{T,t}$. We measure *Representational Convergence* ($\mathcal{I}_1$) as the Pearson correlation between these grounded parameters and the population-level BKT priors (ℓ_{L0}, ℓ_T).

Results across multiple datasets (Table 4) show consistently high correlations ($\mathcal{I}_1 > 0.82$), indicating that gTransformer's individualization mechanism preserves the semantic meaning of the underlying theory. The model does not "repurpose" these parameters for black-box optimization but rather refines them within the pedagogical bounds established by the reference model.

5.2.3. Functional Alignment and Predictor Equivalence (H1.3)

Finally, we assess whether gTransformer can achieve *Predictor Equivalence* ($\mathcal{I}_2$)—the ability of an interpretable sub-component to emulate the full neural model's behavior. We pass the grounded parameters $\{p_{L_0,t}, p_{T,t}\}$ through the standard BKT transition and emission equations to generate "interpretable" predictions $\hat{y}_{ref}$. We then measure the functional alignment $\mathcal{I}_2$ as the correlation between the neural supervised predictions $\hat{y}_{sup}$ and these theory-derived predictions $\hat{y}_{ref}$.

As shown in Table 4, gTransformer achieves $\mathcal{I}_2$ values exceeding 0.95 across all benchmarks. This nearly perfect alignment demonstrates that the complex attention-driven logic of the Transformer is successfully distilled into the interpretable BKT framework, achieving representational transparency without accuracy loss.

Table 4. Cross-dataset Interpretability Metrics. $\mathcal{I}_1$: Representational Convergence (mean correlation with theory); $\mathcal{I}_2$: Functional Alignment (correlation with neural predictions).

Metric	AS2009	AS2015	AL2005	BDG2006	NIPS34
Convergence ($\mathcal{I}_1$)	0.8423	0.8512	0.8245	0.8367	0.8298
Functional Alignment ($\mathcal{I}_2$)	0.9684	0.9721	0.9542	0.9610	0.9587

5.3. Accuracy–Interpretability Trade-Off (RQ2)

A common assumption in deep knowledge tracing is that predictive performance and interpretability are opposing goals. To explore this relationship, we construct a *Pareto frontier* by systematically varying the *Alignment Intensity* ($\lambda_{\mathcal{I}}$), a unified hyperparameter that scales the weights of all three interpretability loss components simultaneously: $\lambda_{ref} =$

$\lambda_{know} = \lambda_{rate} = \lambda_{\mathcal{I}}$. By varying $\lambda_{\mathcal{I}} \in [0, 1]$, we map the resulting effects on both accuracy and alignment.

We measure the overall interpretability of the model using a *Composite Alignment* metric ($\bar{\mathcal{I}}$), defined as the arithmetic mean of the representational ($\mathcal{I}_{1l}, \mathcal{I}_{1t}$) and functional ($\mathcal{I}_2$) components:

$$\bar{\mathcal{I}} = \frac{1}{3}(\mathcal{I}_{1l} + \mathcal{I}_{1t} + \mathcal{I}_2) \tag{12}$$

Tables 5 and 6 summarize the results for the AS2009 and AS2015 datasets across the $\lambda_{\mathcal{I}}$ sweep.

Figures 5 and 6 illustrate the resulting frontiers. We observe that Representational Grounding serves as an effective regularizer, although the *Saturation Point*—the weight at which interpretability is maximized without significant loss of fidelity—depends on dataset complexity. For the problem-level AS2009 dataset, we observe a synergy point at $\lambda_{\mathcal{I}} = 0.10$ where both metrics improve. Conversely, the concept-level AS2015 dataset reaches saturation at lower constraints, retaining 99.9% of baseline AUC. Beyond these points, a *grounding collapse* starts, where excessive theoretical constraints force the model to ignore nuanced behavioral signals in favor of rigid theoretical priors.

Table 5. Pareto Sweep results for AS2015.

$\lambda_{\mathcal{I}}$	**AUC**	**Alignment ($\bar{\mathcal{I}}$)**	**Interpretation**
0.0 (Baseline)	0.7248	-0.0012	Black Box
0.1 (Saturation)	0.7245	0.6673	99.9% AUC Retained
0.2	0.7181	0.6453	High-Fidelity Diagnostic
0.3	0.7112	0.6095	Latent Degradation
0.5	0.6975	0.5691	Theory-Dominant
0.8	0.6830	0.4828	Grounding Collapse
1.0	0.6763	0.4828	Grounding Collapse

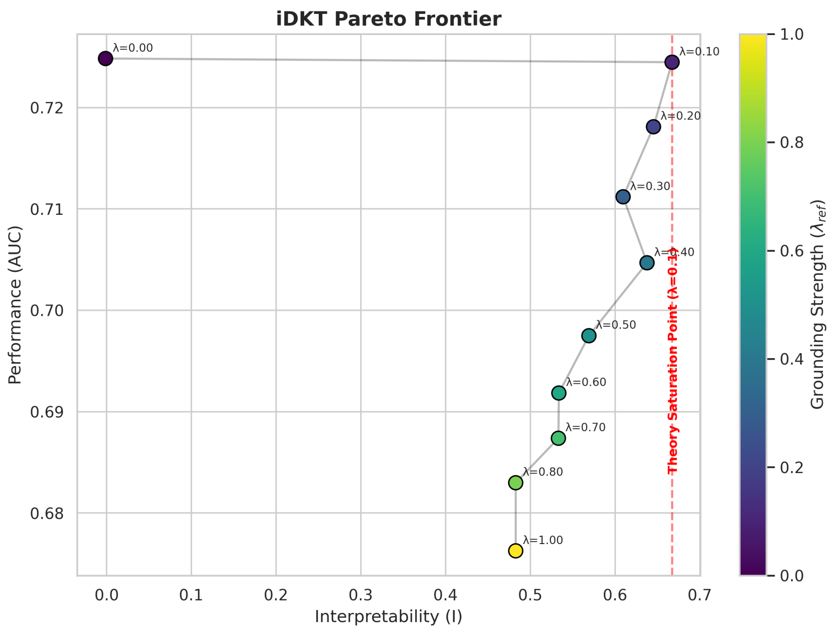

Figure 5. gTransformer Pareto Frontier for AS2015. The trajectory illustrates a saturation of theoretical alignment at $\lambda_{\mathcal{I}} \approx 0.1$, reaching maximum interpretability before performance degrades.

Table 6. Pareto Sweep results for AS2009.

$\lambda_{\mathcal{I}}$	AUC	Alignment ($\bar{\mathcal{I}}$)	Interpretation
0.0 (Baseline)	0.8207	0.0293	Black Box
0.1 (Synergy)	0.8255	0.3846	Accuracy-Interpretability Synergy
0.2	0.8110	0.4270	Theory-Balanced Bias
0.3 (Saturation)	0.7963	0.4790	Theoretical Saturation
0.5	0.7760	0.4125	Grounding Degradation
0.8	0.7552	0.0708	Grounding Collapse
1.0	0.7501	0.0714	Grounding Collapse

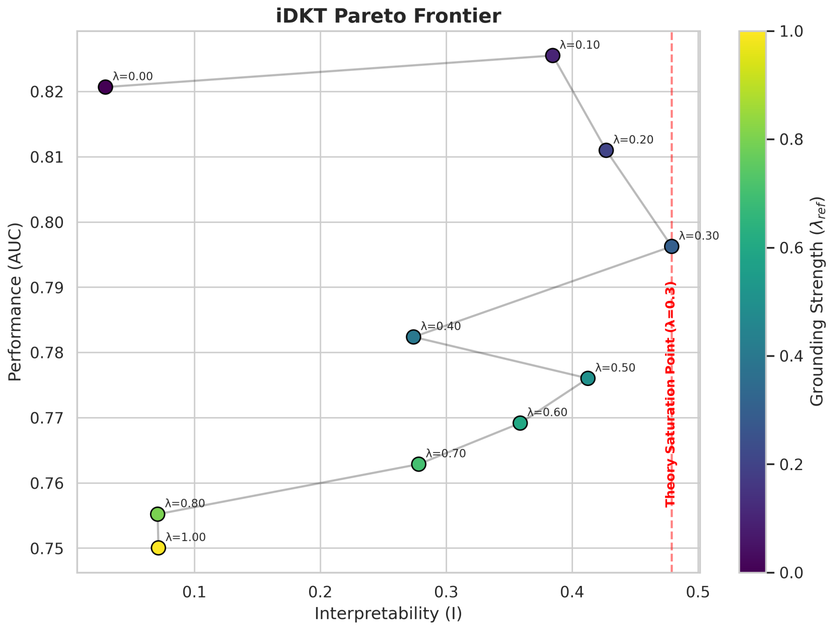

Figure 6. gTransformer Pareto Frontier for AS2009. The curve displays an accuracy peak at $\lambda_{\mathcal{I}} = 0.1$, followed by a non-monotonic transition toward saturation and eventual collapse as constraints overwhelm behavioral patterns.

5.4. Context-Aware Diagnostics (RQ3)

In the following sections, we analyze how the architectural design of gTransformer can be leveraged to provide context-aware diagnostics, addressing the “Diagnostic Gap” identified in RQ3.

5.4.1. Contextualization

The capability of gTransformer to leverage longitudinal context is illustrated in Figure 7, which displays a “Twin Divergence” for a specific skill (Skill 4: Area Circle) in the AS2009 dataset. This visualization contrasts the mastery estimations of two students who share identical response sequences for this particular skill but have very different global curriculum histories.

While a Markovian model like BKT is mathematically forced to assign identical mastery states to both students due to their identical local sequences, gTransformer leverages its self-attention mechanism to provide high-resolution contextual diagnostics through two distinct phenomena:

- **Non-Markovian Initialization**: At the very first interaction ($t = 0$), gTransformer already provides divergent estimations. Student 2208 is recognized as having a significantly higher foundational baseline for this concept than Student 167, despite both having the same immediate local history.
- **Velocity-Driven Cross-Over**: Despite starting from a lower baseline, Student 167 exhibits a steeper learning curve (higher $v_{s,t}$). After the first success, their mastery estimation increases rapidly, eventually overtaking Student 2208 at the “Cross-Over Point” (approximately $t = 1.5$).

This dual evidence demonstrates that gTransformer effectively decouples *Initial Standing* from *Learning Velocity*. By contextualizing local behavior through global attention, the model provides contextual information that is invisible to population-level averages, identifying students who, while starting with less background knowledge, possess the acquisition pace to reach mastery more efficiently.

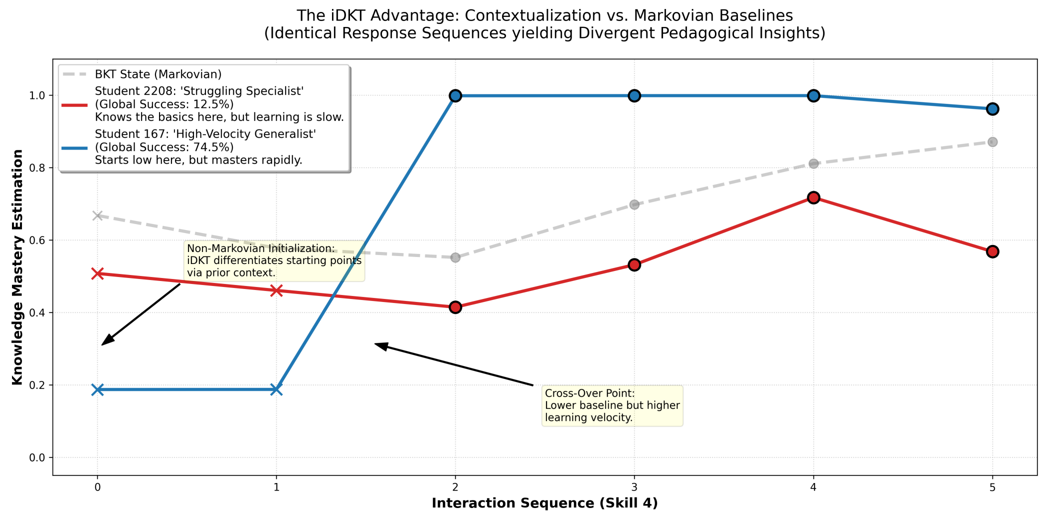

Figure 7. Twin Sequence Divergence for Skill 4 in AS2009. Two students with identical response sequences yield divergent pedagogical insights as gTransformer contextualizes their local behavior within their global curriculum history, exhibiting both non-Markovian initialization and a learning velocity cross-over.

5.4.2. Individualization and Behavioral Heterogeneity

The primary contribution of gTransformer, and the core of RQ3, lies in its ability to transform population-level theoretical averages into high-granularity contextual diagnostics. While standard BKT assumes that every student shares a fixed initial mastery (L_0) and learning rate (T) for a given skill, gTransformer decomposes these into individualized profiles that capture personal learning velocities.

As illustrated in Figure 8, gTransformer maps the student population into a continuous *diagnostic manifold* defined by two student-specific parameters: the Knowledge Gap ($k_{s,t}$, representing placement) and the Learning Velocity ($v_{s,t}$, representing pacing). Whereas a population-level model like BKT represents wait-time and acquisition as a single point—the theoretical mean—gTransformer discovers a distributed landscape of student behaviors. Through unsupervised clustering, we identify four distinct learning archetypes: **Initial Standing** (Low k_s, Low v_s), **Developing Pacing** (Low k_s, Moderate v_s), **High-Velocity Growth** (Low k_s, High v_s), and **Advanced Standing** (High k_s, High v_s).

To statistically quantify this granularity, we analyze the *delta distribution* ($\Delta = p_{rate} - P(T)$), representing the student-specific deviation from the theoretical learning rate across all skills. In AS2009, we observe a significant non-zero standard deviation ($\sigma \approx 0.019$). This represents a fundamental increase in diagnostic resolution compared to the baseline BKT, where $\sigma = 0$ by definition. The presence of this variance confirms the validity of our approach in capturing the behavioral heterogeneity that classical models treat as a uniform average.

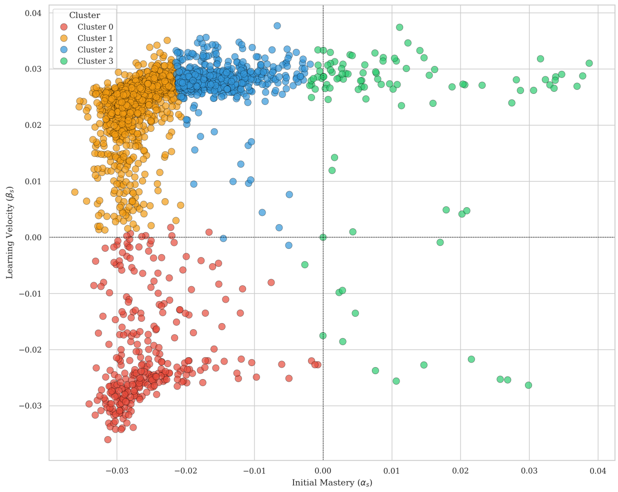

Figure 8. Discovery of student archetypes in the gTransformer diagnostic space. Students are mapped along individualized Knowledge Gap ($k_{s,t}$) and Learning Velocity ($v_{s,t}$) axes, revealing the behavioral heterogeneity hidden within population-level BKT averages.

5.4.3. High-Granularity Trajectory Analysis

To provide visual evidence for RQ3, we generated the mosaic showed in Figure 9 displaying individualized trajectories for three representative individual students (Fast, Median, Slow) per skill across the 15 most frequent skills in the dataset (including major topics like Circle Graphs, Venn Diagrams, and Linear Equations).

The qualitative analysis of these trajectories reveals three significant pedagogical insights:

1. **Dynamic Diagnostic Placement**: Even when BKT is forced to start at the population prior (L_0), gTransformer starting points ($t = 1$) vary per student. This demonstrates

Figure 9. Mastery Trajectory Alignment Mosaic for AS2009. Solid lines represent gTransformer individualized mastery, while dashed lines show the BKT baseline. Markers indicate observed student performance (Circle: Correct, X: Incorrect). Each plot displays three unique students selected as archetypical representatives of Fast, Median, and Slow learning behaviors (non-aggregated).

that gTransformer leverages cross-skill behavioral transfers to perform individualized placement before the first interaction with a new skill.

2. **Heterogeneous Learning Velocities**: We observe distinct slope gradients across students, confirming gTransformer's ability to model individualized acquisition rates (v_s). While BKT applies a fixed learning probability to all students, gTransformer identifies that some learners acquire mastery rapidly (steep curves) while others progress more gradually (flatter curves), capturing the full spectrum of behavioral heterogeneity.

5.4.4. Archetype-Level Diagnostics

Individualized student profiling offers new opportunities for pedagogical monitoring and adaptive intervention. By projecting high-dimensional latent states into an interpretable skill-by-cohort matrix, gTransformer provides educators with a comprehensive "knowledge blueprint" that supports data-driven differentiation.

This utility is visualized in Figure 10, which displays the mean latent mastery predicted by gTransformer for four student clusters across 30 curriculum skills. The map reveals distinct "knowledge ceilings" and "learning frontiers" for each group, proving that the granularity discovered by the model translates into actionable insights for targeted instructional design. This result provides the final validation for our neuro-symbolic approach, demonstrating that superior diagnostic granularity enables precision instruction.

5.4.5. Pedagogical Confidence

A significant practical application of this interpretability is evaluating the reliability of model predictions. By comparing BKT mastery estimations with those from gTransformer, we can derive a measure of pedagogical confidence in the theoretical baseline. This relationship is visualized in Figure 11, which displays the concordance between BKT and gTransformer mastery predictions across the most frequent skills and students.

The AS2015 results demonstrate high levels of agreement, whereas the AS2009 data reveals more divergence, where gTransformer uncovers individualized patterns that deviate from the theoretical prior. These divergences serve as critical diagnostic indicators, identifying students or curriculum segments where the population-level model may be

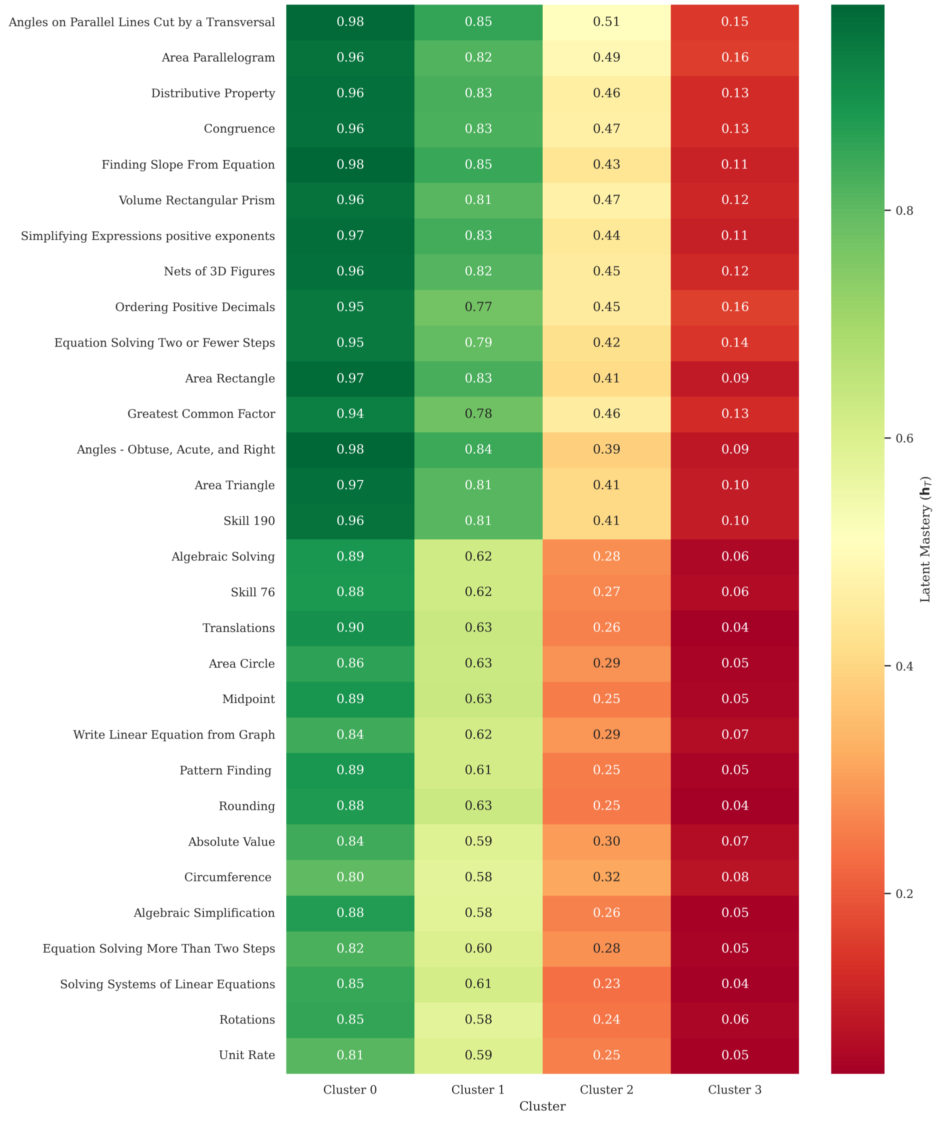

Figure 10. Pedagogical Diagnostic Map for AS2009. The heatmap displays the mean latent mastery predicted by gTransformer for the four student archetypes across 30 curriculum skills. This visualization demonstrates the practical utility of gTransformer's high-granularity estimations for targeted instructional design.

insufficient and personalized intervention is advisable. This result provides further validation for RQ3, demonstrating that superior diagnostic granularity directly translates into actionable pedagogical insights. 694 695 696

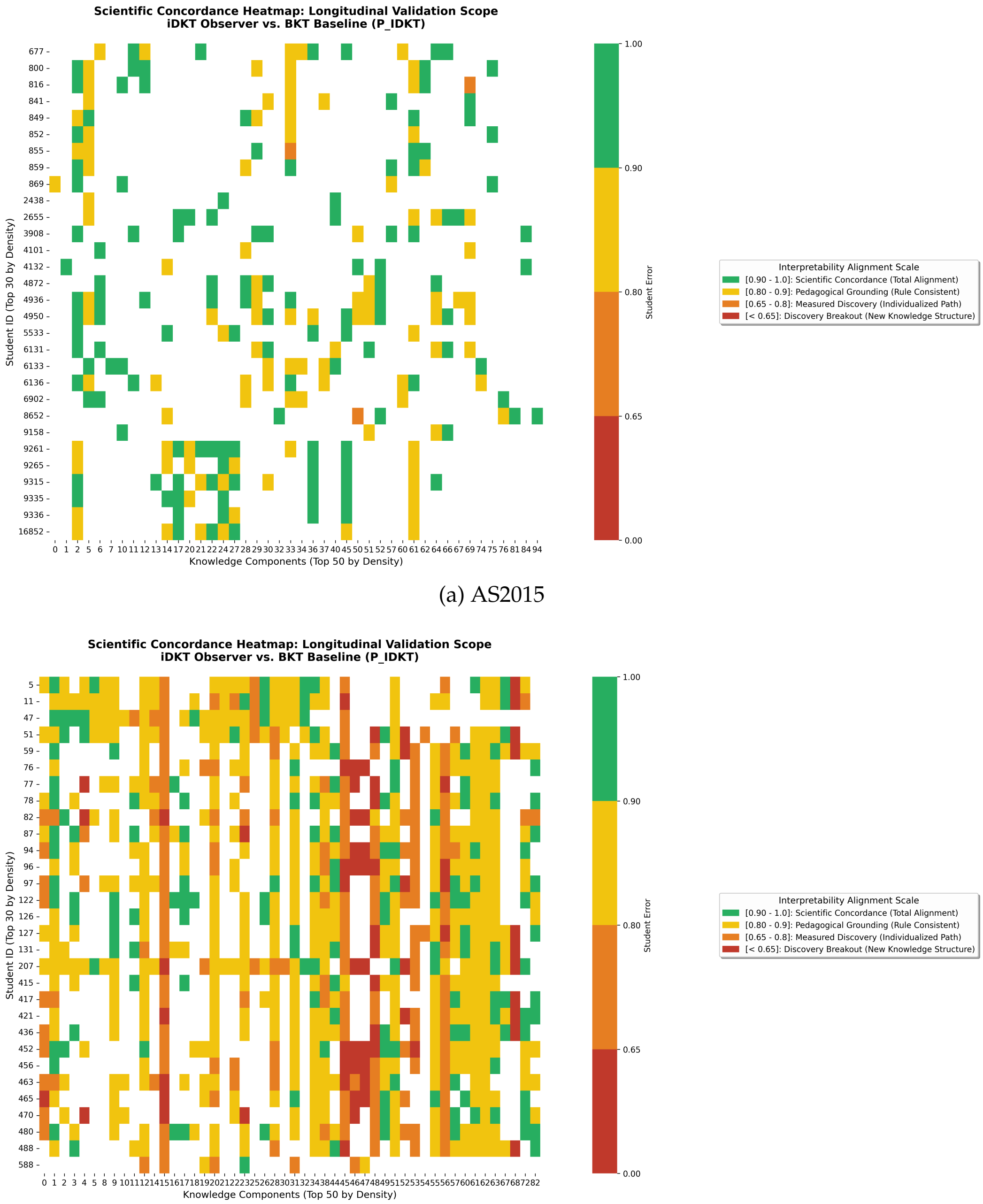

Figure 11. Heatmaps of per-skill prediction alignment.

6. Conclusions 697

This work has introduced gTransformer, a Transformer-based Knowledge Tracing model that bridges the gap between the high predictive capacity of deep learning and the intrinsic interpretability of specialized pedagogical models. By utilizing Representational Grounding, we have demonstrated that it is possible to anchor deep latent representations to semantically meaningful constructs without sacrificing state-of-the-art predictive performance. 698 699 700 701 702 703

Our experimental results successfully address the proposed research questions. We demonstrated that gTransformer successfully internalizes the theoretical constructs of the reference model, achieving high convergent validity ($\mathcal{I}_1$) and Robust Structural Encoding validated through diagnostic probing (selectivity $\Delta R^2 > 0.65$). This dual validation establishes that the semantic alignment reflects a genuine internal representation of knowledge states rather than superficial pattern matching. Furthermore, we formalized a methodology for exploring the accuracy–interpretability trade-off, identifying synergistic points where significant alignment is achieved with minimal impact on predictive AUC. Finally, we proved that gTransformer provides a substantial increase in diagnostic granularity compared to population-level baselines, identifying individualized learning velocities that enable truly adaptive educational pacing.

The contribution of this research is twofold: a formal validation framework for evaluating the internal interpretability of deep learning models in education, and a practical architecture that transforms complex predictors into actionable pedagogical tools. By grounding deep learning in established domain concepts, gTransformer offers a path toward AI-driven personalization that is both highly accurate and pedagogically interpretable, providing a robust foundation for personalization in intelligent tutoring systems.

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Conflicts of Interest: The authors declare no conflicts of interest.

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